Air Freight Multi-Commodity Routing

MVP Formal Model — Phase 1, v1

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Draft — 2026-05-16

Abstract

Formal specification of the optimization model for the air freight multi-commodity routing problem. The model is a Binary Multi-Commodity Flow problem on a commodity-specific subgraph of the air freight network (see `docs/air_freight_multi_shipment_graph.drawio` and `docs/air_freight_shipment_subgraph.drawio`). Scope: single-shipment routing across multiple flight legs (direct and multi-hop) with two air capacity layers — contracted ULD allocation (Block Space Agreements) and spot rate-card pricing. Probabilistic transit time objectives, through-ULD as a decision variable, period-level utilization commitments, and full bin-packing inside ULDs are deferred. The model parallels the Ocean FCL formulation (`model/ocean_fcl_routing.tex`) for notational consistency.

Contents

1 Problem Statement

Given a set of air shipment requests (commodities), each defined by origin location, destination location, cargo volume, chargeable weight, and time constraints, find a minimum-cost assignment of commodities to paths through the multimodal air freight network such that:

- every commodity is delivered on or before its deadline,
- no flight leg exceeds its ULD physical capacity (volume and weight per ULD),
- no Block Space Agreement (BSA) exceeds its contracted per-flight ULD allocation,
- period-level allocation caps are respected,
- cargo cutoff times are honored,
- hub minimum connect times are honored, including the through-ULD vs. re-ULD distinction, and
- optional per-commodity budget caps are respected.

MVP scope.

- Single-shipment routing; each shipment routes independently subject to shared supply.
- Each scheduled flight leg is one arc. Multi-stop flights (same flight number, intermediate stops) are decomposed into multiple arcs for transit time visibility.
- ULD types: LD3, LD7, PMC pallet, AKE. Optimizer selects ULD type per shipment per flight.
- Two air supply layers per flight: (a) contracted ULD allocation under one or more BSAs; (b) spot rate-card at IATA weight breaks.
- Pivot weight (take-or-pay minimum) modeled on hard BSA contracts.
- Through-ULD vs. re-ULD at hub modeled via pre-computed compatibility parameter ψ, with rehandling cost and MCT differentiated. Explicit decision variable for through-ULD selection deferred to P1.
- Period-level minimum utilization commitments (hard BSA take-or-pay across period) are computed by a separate upstream model and surfaced as fixed per-run targets. This MILP does not solve them jointly.
- Objective: minimize total air + ground freight cost subject to deterministic time-window feasibility. Probabilistic on-time delivery objective deferred to P1.
- Hazmat / perishable / valuable cargo type segregation: model parameter, not a hard MVP constraint (recorded for future restriction logic).

2 Graph Structure

2.1 Physical Network Graph $G(N, A)$

The air freight network is a directed graph $G = (N, A)$.

$$N = N_O \cup N_{\text{CFS,O}} \cup N_{\text{AP,O}} \cup N_{\text{AP,H}} \cup N_{\text{AP,D}} \cup N_{\text{CFS,D}} \cup N_D$$

N_O — **Origins** One per shipper origin door (e.g., factory, warehouse). Commodity source nodes.

$N_{\text{CFS,O}}$ — **Origin CFS**
Container Freight Stations near origin airports (e.g., TPE-CFS, KHH-CFS). ULD build-up location. Airline-owned empty ULDs are delivered here;

cargo from one or more shipments is stuffed into a ULD; built ULD is trucked to the airline cargo terminal.

$N_{\mathbf{AP,O}}$ — **Origin airports**
Airline cargo terminals at departure airports (e.g., TPE, KHH). Built ULDs are received here for loading onto aircraft.

$N_{\mathbf{AP,H}}$ — **Hub airports**
Intermediate airports on multi-hop routes (e.g., HKG, NRT, ANC). Through-cargo handling or re-ULDing occurs here.

$N_{\mathbf{AP,D}}$ — **Destination airports**
Airline cargo terminals at arrival airports (e.g., JFK, ORD, LAX). ULDs are unloaded here.

$N_{\mathbf{CFS,D}}$ — **Destination CFS**
CFS near destination airport for ULD breakdown. Trucked from airline cargo terminal; ULDs broken down; individual consignments released for final delivery.

N_D — **Destinations**
One per consignee delivery door. Commodity sink nodes.

$$A = A_{\text{PU}} \cup A_{\text{OC}} \cup A_{\text{AIR}} \cup A_{\text{DC}} \cup A_{\text{FD}}$$

$A_{\mathbf{PU}}$ — **Pickup** (i, j), $i \in N_O$, $j \in N_{\text{CFS,O}}$. Truck pickup from origin door to origin CFS.

$A_{\mathbf{OC}}$ — **Origin CFS to origin airport**
(i, j), $i \in N_{\text{CFS,O}}$, $j \in N_{\text{AP,O}}$. Built ULDs trucked from CFS to airline cargo terminal. Includes fixed CFS build-up time as part of effective arc duration.

$A_{\mathbf{AIR}}$ — **Flight legs**
(i, j), $i, j \in N_{\text{AP,O}} \cup N_{\text{AP,H}} \cup N_{\text{AP,D}}$. One arc per scheduled flight leg (specific carrier, flight number, departure date, origin airport, destination airport). A multi-stop flight (e.g., f: TPE→ANC→JFK) decomposes into two air arcs sharing the same flight number metadata.

$A_{\mathbf{DC}}$ — **Destination airport to destination CFS**
(i, j), $i \in N_{\text{AP,D}}$, $j \in N_{\text{CFS,D}}$. Trucked from airline cargo terminal to CFS. Includes fixed CFS breakdown time.

$A_{\mathbf{FD}}$ — **Final delivery**
(i, j), $i \in N_{\text{CFS,D}}$, $j \in N_D$. Trucked from destination CFS to consignee door.

Remark 1 (Origin and destination CFS as explicit nodes)**.** Unlike ocean (where the POL terminal is one node), air introduces a mandatory CFS step at origin (ULD build-up) and destination (ULD breakdown). Modeling CFS as an explicit node permits: (a) the fixed CFS dwell time to enter feasibility checks, (b) future modeling of CFS dock capacity, ULD availability at CFS, and (c) alternative-CFS routing (e.g., a Kaohsiung shipment trucked to TPE-CFS for consolidation).

Remark 2 (Flight legs vs. flight numbers)**.** A real-world flight number (e.g., 5Y9123 TPE→ANC→JFK) operates one aircraft sortie with an intermediate stop. We model this as *two arcs* in A_{AIR} — (TPE, ANC) and (ANC, JFK) — both tagged with the same flight number. This preserves transit time visibility at each leg and provides the structural hook for per-leg ULD allocation accounting. The fact that the ULD physically stays on the aircraft across the intermediate stop is captured by the through-ULD compatibility parameter ψ (Section 4.8).

Figure reference. See the rendered network diagrams: `docs/air_freight_multi_shipment_graph.drawio` (multi-shipment view, supply sharing) and `docs/air_freight_shipment_subgraph.drawio` (single-shipment subgraph $G(N_k, A_k)$ for S3 TPE→NYC). Render the .drawio sources to PDF via drawio export before final approval.

2.2 Commodity-Specific Subgraph $G(N_k, A_k)$

The MILP for commodity k operates on a subgraph $G(N_k, A_k) \subseteq G(N, A)$ containing only arcs that (i) are time-feasible for commodity k, (ii) lie on at least one complete path from $o(k)$ to $d(k)$, and (iii) are compatible with the commodity's cargo type and ULD constraints (e.g., hazmat-restricted flights excluded for non-DGR-certified carriers). See Section 5 for the construction algorithm.

3 Sets and Indices

Symbol	Description
K	Set of commodities (shipments), indexed by k
N, A	All nodes and arcs
A_{PU}, A_{OC}, A_{AIR}, A_{DC}, A_{FD}	Arc subsets by type
$A^k \subseteq A$	Feasible arc set for commodity k (Section 5)
$A^k_S \triangleq A^k \cap A_S$	Feasible arcs of type S for commodity k
N_k	Nodes incident to A^k
F	Set of scheduled flight legs, indexed by f. Each flight is one arc in A_{AIR}.
$F^k \subseteq F$	Flights corresponding to arcs in A^k_{AIR}
$f(i, j)$	Mapping: air arc $(i, j) \in A_{\text{AIR}}$ to its flight $f \in F$
U	Set of ULD types: $U = \{\text{LD3, LD7, PMC, AKE}\}$
$U_f \subseteq U$	ULD types available on flight f (depends on aircraft type)
C	Set of ULD allocation contracts (BSAs), indexed by c
$C_f \subseteq C$	Contracts that allocate ULDs on flight f
T	Set of allocation time periods (e.g., monthly), indexed by t
$\tau_k(i)$	Earliest arrival time of commodity k at node i
$\text{Hub}(j)$	Set of consecutive air-arc pairs at hub node $j \in N_{\text{AP,H}}$ for shipment k: $\{((i, j), (j, \ell)) : (i, j), (j, \ell) \in A^k_{\text{AIR}}\}$

4 Parameters

4.1 Commodity Parameters

Parameter	Symbol	Description	Unit
Volume	v_k	Cargo volume (sum of all pieces)	CBM
Actual weight	w_k	Gross weight	kg
Chargeable weight	cw_k	$\max(w_k,\ v_k \cdot 167)$	kg
Piece dimensions	D_k	Max single-piece L×W×H (for ULD fit check)	cm
Cargo ready	t_k^{rdy}	Earliest pickup time from origin door	datetime
Deadline	T_k^{dead}	Latest acceptable arrival at destination	datetime
Origin	$o(k)$	Source node $\in N_O$	—
Destination	$d(k)$	Sink node $\in N_D$	—
Cargo type	τ_k	General / DGR / PER / VAL / HEA	categorical
HS code	hs_k	6-digit Harmonized System code	—
Incoterm	inco_k	EXW / FCA / CIF / DDP	—
Budget cap	B_k	Hard cost ceiling; ∞ if not set	USD

Chargeable weight.

$$cw_k \triangleq \max(w_k,\ v_k \cdot 167) \tag{1}$$

IATA standard: $1\,\mathrm{m}^3 = 167\,\mathrm{kg}$ volumetric equivalent. Dense cargo (machinery, electronics) rated on w_k; bulky-light cargo (furniture, packaging) rated on volumetric.

4.2 Ground Arc Parameters (Pickup, CFS, Final Delivery)

For each ground arc $(i,j) \in A_{\mathrm{PU}} \cup A_{\mathrm{OC}} \cup A_{\mathrm{DC}} \cup A_{\mathrm{FD}}$:

Parameter	Symbol	Description	Unit
Cost	c_{ij}^{G}	Truck cost per shipment	USD
Mean transit	μ_{ij}^{G}	Mean transit time	days
Std dev	σ_{ij}^{G}	Std dev of transit	days
CFS build-up time	β^{O}	Fixed origin CFS dwell (ULD build-up). Configurable.	hours
CFS breakdown time	β^{D}	Fixed destination CFS dwell (ULD breakdown). Configurable.	hours

The effective transit time on the CFS→airport arc $(i,j) \in A_{\mathrm{OC}}$ includes the CFS build-up: $\tilde{\mu}_{ij}^{\mathrm{OC}} = \mu_{ij}^{\mathrm{OC}} + \beta^{\mathrm{O}}$. Likewise for A_{DC} with β^{D}.

4.3 Flight (Air Arc) Parameters

For each flight $f \in F$ (equivalently, each $(i,j) \in A_{\mathrm{AIR}}$):

Parameter	Symbol	Description	Unit
Carrier	carrier(f)	IATA carrier code (CI, BR, CX, JL, NH, 5Y, ...)	—
Flight number	fno(f)	Flight number; multiple legs may share a number	—
Aircraft type	ac(f)	777F, 747-8F, A330F, 777-300ER, A350-900, ...	—
ETD	ETD_f	Scheduled departure datetime	datetime
ETA	ETA_f	Scheduled arrival datetime	datetime
Block time	μ_f^{AIR}	$\mathrm{ETA}_f - \mathrm{ETD}_f$	hours
Std dev	σ_f^{AIR}	Std dev of actual block time	hours
Cargo cutoff	CGC_f	Latest time ULDs must be tendered to airline cargo terminal	datetime
DGR certified	dgr(f)	Accepts dangerous goods?	boolean
Cold chain	per(f)	Maintains $\leq 8°$C?	boolean

4.4 ULD Specifications

ULD type	Code	Volume	Payload	Aircraft compatibility
LD3	AKE/AVE	4.5 m^3	1,587 kg	Wide-body belly + freighter
LD7	ALE/AAY	11.1 m^3	4,626 kg	Wide-body belly + freighter
PMC pallet	PMC/PAG	7.5 m^3	6,804 kg	Freighter main deck
AKE small	AKE	4.5 m^3	1,497 kg	Wide-body belly

For each $u \in U$:

Parameter	Symbol	Description	Unit
Volume capacity	V_u	ULD usable volume (Table above)	CBM
Weight capacity	W_u	ULD payload limit	kg
Fill rate cap	η	Max fraction of CBM usable in practice (default 0.97)	—

For each flight f and ULD type $u \in U_f$:

Parameter	Symbol	Description	Unit
Aircraft ULD positions	$P_{f,u}$	Number of u-type ULD positions on ac(f)	—

$P_{f,u}$ is an upper bound on physical positions for ULD type u on the aircraft. It is independent of the forwarder's contracted allocation.

4.5 Contracted ULD Allocation (Block Space Agreement) Parameters

A Block Space Agreement (BSA) reserves a fixed allocation of ULD positions per flight at a contracted rate. For each contract $c \in C$:

Parameter	Symbol	Description	Unit
Flights covered	$F_c \subseteq F$	Set of scheduled flights covered by contract c	—
Per-flight allocation	$\text{alloc}(c, f, u)$	ULD count of type u allocated on flight f under contract c	ULDs
Period allocation	$\text{alloc}(c, u, t)$	Total ULDs of type u allocated in period t under contract c	ULDs
Period utilization	$\text{util}(c, u, t)$	Already-booked ULDs in period t at solve time	ULDs
Remaining (period)	$\text{rem}(c, u, t)$	$\text{alloc}(c, u, t) - \text{util}(c, u, t)$	ULDs
Min utilization (period)	$\underline{M}_{c,u,t}$	Hard BSA minimum: external upstream model output. Default 0 for soft BSA.	ULDs
Pivot weight	$\pi_{c,u}$	Minimum chargeable weight per ULD per flight under c	kg/ULD
Contract rate	r_c	Contracted price	USD/kg chargeable
Hard vs. soft	hard_c	Boolean: hard BSA (take-or-pay) or soft (cancellable)	—

Remark 3 (Period commitments are upstream)**.** $\underline{M}_{c,u,t}$ is computed by a separate upstream model that allocates how much of each hard BSA's monthly take-or-pay commitment should be met by each routing run within the period. The routing MILP takes $\underline{M}_{c,u,t}$ as a fixed input per run. The MILP does *not* simultaneously optimize period-level commitment allocation; that is a higher-level decision.

Remark 4 (Pivot weight semantics)**.** The pivot $\pi_{c,u}$ is the minimum chargeable weight per ULD per flight regardless of actual cargo loaded. If a forwarder claims one LD3 with pivot 1,000 kg on a flight but loads only 400 kg of cargo, the forwarder pays for 1,000 kg $\cdot\, r_c$. This is the take-or-pay floor on hard BSAs.

4.6 Spot Rate-Card Parameters

For each flight f, the airline offers spot rate-card pricing with IATA weight breaks:

Parameter	Symbol	Description	Unit
Break tiers	$B = \{N, 45, 100, 300, 500, 1000\}$	IATA standard break points (kg)	kg
Tier rate	$r^{\text{sp}}_{f,b}$	Rate per kg in tier b on flight f	USD/kg
Tier minimum	$b_{\min}$	N tier: minimum charge floor	USD

Spot rate cost per shipment. For shipment k tendered on flight f at the spot rate, the cost is the minimum across applicable tiers:

$$\text{cost}^{\text{spot}}(k, f) \triangleq \min_{b \in B} r^{\text{sp}}_{f,b} \cdot \max(cw_k,\, b) \tag{2}$$

Since cw_k is a parameter (known at planning time), $\text{cost}^{\text{spot}}(k, f)$ is pre-computable. The IATA rule that "charging at the next tier with a lower rate may yield a lower total cost" is captured automatically by the min.

4.7 Surcharge Parameters

Per shipment per flight, the surcharge stack adds:

Parameter	Symbol	Description	Unit
Fuel surcharge	FSC_f	USD/kg, monthly refresh	USD/kg
Security surcharge	SSC_f	USD/kg	USD/kg
Origin THC	THC_f^O	Per shipment, origin terminal handling	USD
Destination THC	THC_f^D	Per shipment, destination terminal handling	USD
AMS fee (US imports)	AMS_f	Per shipment	USD

4.8 Hub Connection Parameters (Through-ULD vs. Re-ULDing)

For each pair of consecutive air arcs (f, g) at hub j — meaning f ends at j and g starts at j — and each ULD type u:

Parameter	Symbol	Description	Unit
Through-ULD compatible	$\psi_{f,g,u}$	1 if a ULD of type u can transit hub j without being broken down (same aircraft for through-flights; or same airline through-cargo agreement)	boolean
MCT (through)	$\mathrm{MCT}_{f,g}^{\mathrm{thru}}$	Min connect time when through-ULD is used (technical-stop turnaround or hub through-cargo time)	hours
MCT (re-ULD)	$\mathrm{MCT}_{f,g}^{\mathrm{reULD}}$	Min connect time when ULD must be broken down and rebuilt at hub	hours
Rehandling cost	$\rho_{f,g,u}^{\mathrm{reULD}}$	Cost per ULD to break down at hub and rebuild for onward flight	USD/ULD

Definition 1 (Re-ULDing)**.** *Re-ULDing* is the operation of breaking down a built ULD at an intermediate (hub) airport and stuffing its cargo into a different ULD for the onward flight. It occurs when:

(a) the two consecutive flights are operated by *different airlines* (interline routing) — each airline owns a distinct ULD pool and ULDs cannot transfer;
(b) the two flights require *different ULD types* (e.g., LD3 on a passenger belly leg and PMC main-deck pallet on a freighter leg);
(c) the hub lacks through-cargo facilities for this ULD type (rare at major hubs).

Re-ULDing is operationally expensive: it adds $\sim$ 4–8 hours of dwell time (breakdown + storage + rebuild), incurs ~\$150–\$500 in handling fees per ULD, and introduces cargo damage risk. When $\psi_{f,g,u} = 1$, the ULD physically stays intact across the hub — either remaining on the same aircraft (through-flight technical stop) or moving as a complete unit between aircraft at hub through-cargo facilities.

MVP simplification. In MVP, $\psi_{f,g,u}$ is a *pre-computed parameter*, not a decision variable. For each consecutive arc pair at a hub:

- If carrier(f) = carrier(g) *and* fno(f) = fno(g) (through-flight, same aircraft, intermediate stop): $\psi_{f,g,u} = 1$ for all $u \in U_f \cap U_g$.
- If carrier(f) = carrier(g) *and* the hub has a through-cargo agreement for type u: $\psi_{f,g,u} = 1$.
- Otherwise (interline or incompatible ULD type): $\psi_{f,g,u} = 0$, re-ULDing required.

The MILP uses ψ to select the applicable MCT and rehandling cost on each used arc pair. The choice between "use through-ULD" and "re-ULD anyway" is not a free decision in MVP; through-ULD is used whenever $\psi = 1$. P1 extension: promote ψ to a decision variable to allow forwarder-elected re-ULDing (e.g., for re-sorting cargo destined to different onward routes at the hub).

5 Commodity-Specific Subgraph Construction

Before the MILP is built, we construct $G(N_k, A_k)$ for each commodity k. The subgraph contains only arcs that are time-feasible, cargo-type-compatible, and reachable on a complete $o(k) \to d(k)$ path.

Construction algorithm.

1. **Pickup pass.** For each origin CFS $j \in N_{\text{CFS,O}}$, compute earliest CFS arrival:
$$\tau_k(j) \triangleq t_k^{\text{rdy}} + \mu_{o(k),j}^{\text{PU}}$$
Arc $(o(k), j) \in A_{\text{PU}}$ is candidate iff there exists a downstream feasible path.
2. **Origin CFS pass.** For each CFS-to-airport arc $(j, a) \in A_{\text{OC}}$ where $j \in N_{\text{CFS,O}}$ and $a \in N_{\text{AP,O}}$, compute earliest airport arrival:
$$\tau_k(a) \triangleq \tau_k(j) + \mu_{j,a}^{\text{OC}} + \beta^{\text{O}}$$
3. **Flight reachability pass.** For each flight $f \in F$ (arc $(a, b) \in A_{\text{AIR}}$): f is candidate for k iff
$$\begin{aligned}
&\tau_k(a) \le \text{CGC}_f && \text{(can make the cargo cutoff)}\\
&\text{ETD}_f + \mu_f^{\text{AIR}} \le T_k^{\text{dead}} - \text{(downstream legs)} && \text{(deadline reachable)}\\
&\tau_k = \text{General or dgr}(f) = 1 \text{ if } \tau_k = \text{DGR} && \text{(cargo-type compatibility)}\\
&\exists\, u \in U_f : D_k \text{ fits } u && \text{(physical fit in some ULD)}
\end{aligned}$$
4. **Hub transitions pass.** For each pair of consecutive air arcs (f, g) at hub h: compute $\tau_k(g_{\text{start}}) = \text{ETA}_f + \text{MCT}^*_{f,g}$ where MCT^* uses MCT^{thru} if $\psi_{f,g,u} = 1$ for some $u \in U_f \cap U_g$, else $\text{MCT}^{\text{reULD}}$.
5. **Destination CFS & delivery pass.** For each destination airport $b \in N_{\text{AP,D}}$ reachable: include arc $(b, c) \in A_{\text{DC}}$ where $c \in N_{\text{CFS,D}}$, then $(c, d(k)) \in A_{\text{FD}}$.
6. **Reachability sweep.** Forward BFS from $o(k)$ and backward BFS from $d(k)$. Retain only arcs on paths reachable from both ends.

6 Decision Variables

Variable	Type	Description	Domain
x_{ij}^k	Binary	1 if shipment k routed on arc $(i,j) \in A^k$	$\{0,1\}$
$y_{f,u,k}^c$	Binary	1 if shipment k uses one ULD of type u on flight f under contract c	$\{0,1\}$
$z_{f,u}^c$	Integer	Number of ULDs of type u claimed on flight f under contract c	$\mathbb{Z}_{\geq 0}$
$b_{f,k}$	Binary	1 if shipment k tendered to flight f at spot rate (not via contract)	$\{0,1\}$
$C_{f,u,c}$	Continuous	Effective chargeable weight billed for contract c, flight f, ULD type u (max of actual and pivot)	$\mathbb{R}_{\geq 0}$
$t_k(i)$	Continuous	Arrival time of shipment k at node $i \in N_k$	$\mathbb{R}_{\geq 0}$

7 Constraints

P.1 — Flow Conservation

For each commodity k and each node $n \in N_k$:

$$\sum_{j:(n,j)\in A^k} x_{nj}^k - \sum_{i:(i,n)\in A^k} x_{in}^k = \begin{cases} +1 & \text{if } n = o(k) \\ -1 & \text{if } n = d(k) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

P.2 — ULD Volume Capacity (per flight, per ULD type, per contract)

For each flight $f \in F$, each ULD type $u \in U_f$, each contract $c \in C_f$:

$$\sum_{k\in K} v_k \cdot y_{f,u,k}^c \;\leq\; \eta \cdot V_u \cdot z_{f,u}^c \tag{4}$$

Total volume loaded into contract-c ULDs of type u on flight f cannot exceed the fill-rate-capped capacity of the claimed ULDs.

P.3 — ULD Weight Capacity

For each f, u, c as above:

$$\sum_{k\in K} cw_k \cdot y_{f,u,k}^c \;\leq\; W_u \cdot z_{f,u}^c \tag{5}$$

P.4 — Per-Flight Contract Allocation Cap

For each flight $f \in F$, ULD type u, contract $c \in C_f$:

$$z_{f,u}^c \;\leq\; \text{alloc}(c, f, u) \tag{6}$$

Forwarder cannot claim more ULDs of type u on flight f than the contract allocates.

P.5 — Aircraft Physical Position Cap (across all contracts)

For each flight $f \in F$, ULD type $u \in U_f$:

$$\sum_{c \in C_f} z^c_{f,u} \;\le\; P_{f,u} \tag{7}$$

Total ULDs of type u across all contracts cannot exceed the aircraft's physical positions.

P.6 — Period Contract Allocation Cap

For each contract c, ULD type u, period t:

$$\sum_{f \in F_c \cap t} z^c_{f,u} \;\le\; \mathrm{rem}(c, u, t) \tag{8}$$

Cumulative ULDs claimed in period t under contract c cannot exceed the period's remaining allocation (after prior bookings).

P.7 — Period Minimum Utilization (hard BSA take-or-pay)

For each contract c with $\mathrm{hard}_c = 1$, ULD type u, period t:

$$\sum_{f \in F_c \cap t} z^c_{f,u} \;\ge\; \underline{M}_{c,u,t} \tag{9}$$

$\underline{M}_{c,u,t}$ is the per-run minimum allocation passed in from the upstream period-commitment model. For soft BSAs, $\underline{M}_{c,u,t} = 0$.

P.8 — Arc-to-ULD Linkage

For each shipment k, flight f with $(i, j) = f^{-1}$ in A^k_{AIR}:

$$x^k_{ij} \;=\; \sum_{c \in C_f} \sum_{u \in U_f} y^c_{f,u,k} \;+\; b_{f,k} \tag{10}$$

If shipment k uses arc (i, j), it is loaded via exactly one contract ULD or the spot rate.

P.9 — Spot vs. Contract Exclusivity (per shipment per flight)

Implicit in P.8: at most one of $\{y^c_{f,u,k}\}_{c,u} \cup \{b_{f,k}\}$ equals 1 when $x^k_{ij} = 1$. Formally:

$$\sum_{c \in C_f} \sum_{u \in U_f} y^c_{f,u,k} \;+\; b_{f,k} \;\le\; 1 \tag{11}$$

P.10 — Effective Chargeable Weight (pivot weight linearization)

For each contract c, flight f, ULD type u — linearizes $\max(\text{actual}, \pi \cdot z)$:

$$C_{f,u,c} \;\ge\; \sum_{k \in K} cw_k \cdot y^c_{f,u,k} \qquad \text{(at least the actual loaded weight)} \tag{12}$$

$$C_{f,u,c} \;\ge\; \pi_{c,u} \cdot z^c_{f,u} \qquad \text{(at least the pivot floor)} \tag{13}$$

Since $C_{f,u,c}$ is minimized in the objective (it multiplies the rate r_c), at optimum $C_{f,u,c} = \max(\sum_k cw_k \cdot y, \pi_{c,u} \cdot z)$ as required. See Section 9 for the linearization argument.

P.11 — Cargo Cutoff

For each shipment k and each air arc $(i,j) \in A^k_{\mathrm{AIR}}$ with $f = f(i,j)$:

$$t_k(i) \;\le\; \mathrm{CGC}_f + M(1 - x^k_{ij}) \tag{14}$$

where M is a big-M constant (use a tight bound, e.g., max planning horizon).

P.12 — Arrival Time Propagation (ground arcs)

For each shipment k and each ground arc $(i,j) \in A^k_{\mathrm{PU}} \cup A^k_{\mathrm{OC}} \cup A^k_{\mathrm{DC}} \cup A^k_{\mathrm{FD}}$:

$$t_k(j) \;\ge\; t_k(i) + \tilde{\mu}^{\mathrm{G}}_{ij} - M(1 - x^k_{ij}) \tag{15}$$

where $\tilde{\mu}^{\mathrm{G}}_{ij} = \mu^{\mathrm{G}}_{ij} + \beta^{\mathrm{O}}$ for A_{OC}, $\mu^{\mathrm{G}}_{ij} + \beta^{\mathrm{D}}$ for A_{DC}, else μ^{G}_{ij}.

P.13 — Arrival Time Propagation (air arcs)

For each shipment k and each air arc $(i,j) \in A^k_{\mathrm{AIR}}$ with $f = f(i,j)$:

$$t_k(j) \;\ge\; \mathrm{ETA}_f - M(1 - x^k_{ij}) \tag{16}$$

P.14 — Hub Minimum Connect Time

For each shipment k and each pair of consecutive air arcs (f,g) at hub h with both $x^k_{(\cdot,h)} = 1$ and $x^k_{(h,\cdot)} = 1$:

$$t_k(g_{\mathrm{start}}) \;\ge\; \mathrm{ETA}_f \;+\; \mathrm{MCT}^*_{f,g} \;-\; M(2 - x^k_f - x^k_g) \tag{17}$$

where $\mathrm{MCT}^*_{f,g} = \mathrm{MCT}^{\mathrm{thru}}_{f,g}$ if $\psi_{f,g,u} = 1$ for the ULD type u used by k on both arcs, otherwise $\mathrm{MCT}^{\mathrm{reULD}}_{f,g}$.

P.15 — Deadline

For each shipment k:

$$t_k(d(k)) \;\le\; T^{\mathrm{dead}}_k \tag{18}$$

P.16 — Cargo Type Compatibility

For each shipment k with $\tau_k = \mathrm{DGR}$ and each air arc $(i,j) \in A^k_{\mathrm{AIR}}$:

$$x^k_{ij} \;\le\; \mathrm{dgr}(f(i,j)) \tag{19}$$

Analogous constraints for perishable, valuable, heavy cargo.

P.17 — ULD Physical Fit

For each shipment k and each (f,u,c) combination where D_k does not fit ULD type u:

$$y^{c}_{f,u,k} \;=\; 0 \tag{20}$$

This is a pre-processing exclusion, not a runtime constraint — fit is determined from D_k vs. ULD contour at subgraph construction. Bin-packing of multiple shipments inside a single ULD is approximated by the volume constraint P.2; full 3D bin-packing is deferred.

P.18 — Budget Cap (optional)

For each shipment k with $B_k < \infty$:

$$\text{cost}_k \;\leq\; B_k \tag{21}$$

where cost_k is the per-shipment cost contribution from the objective.

P.19 — Domain

$$x_{ij}^k, y_{f,u,k}^c, b_{f,k} \in \{0,1\}, \quad z_{f,u}^c \in \mathbb{Z}_{\geq 0}, \quad C_{f,u,c}, t_k(i) \in \mathbb{R}_{\geq 0} \tag{22}$$

8 Objective Function

Minimize total cost across all shipments:

$$\begin{aligned}
\min \quad & \sum_{k\in K} \sum_{(i,j)\in A_{\text{PU}}^k \cup A_{\text{OC}}^k \cup A_{\text{DC}}^k \cup A_{\text{FD}}^k} c_{ij}^{\text{G}} \cdot x_{ij}^k && \text{(ground cost)} \\
& + \sum_{c\in C} \sum_{f\in F_c} \sum_{u\in U_f} r_c \cdot C_{f,u,c} && \text{(contracted ULD cost incl. pivot floor)} \\
& + \sum_{k\in K} \sum_{f\in F^k} \text{cost}^{\text{spot}}(k,f) \cdot b_{f,k} && \text{(spot rate cost)} \\
& + \sum_{k\in K} \sum_{f\in F^k} \big(\text{FSC}_f + \text{SSC}_f\big) \cdot cw_k \cdot \mathbb{1}[k \text{ on } f] \;+\; \text{(THC, AMS — per shipment per arc)} \\
& && \text{(surcharges)} \\
& + \sum_{k\in K} \sum_{(f,g)\in \text{Hub}(k)} \sum_{u} \rho_{f,g,u}^{\text{reULD}} \cdot (1-\psi_{f,g,u}) \cdot y_{f,u,k}^c \cdot y_{g,u,k}^{c'} && \text{(rehandling cost when re-ULDing)}
\end{aligned}$$

The rehandling cost term is bilinear ($y \cdot y$); linearize by introducing an auxiliary binary variable per consecutive air-arc pair per shipment per ULD type that equals the product (Section 9).

Indicator $\mathbb{1}[k$ on $f]$. Equals 1 if shipment k uses flight f; can be substituted as $\sum_{c,u} y_{f,u,k}^c + b_{f,k}$.

9 Linearization Details

9.1 Pivot Weight Linearization (P.10)

The native cost expression per (contract, flight, ULD type) is:

$$\text{cost}_{f,u,c} = r_c \cdot \max\Big(\sum_k cw_k \cdot y_{f,u,k}^c,\; \pi_{c,u} \cdot z_{f,u}^c\Big)$$

The max operator is non-linear. Introduce auxiliary variable $C_{f,u,c} \in \mathbb{R}_{\geq 0}$ with two linear inequality constraints (P.10a, P.10b). Replace the cost term with $r_c \cdot C_{f,u,c}$ in the objective. **Why this works:** the objective minimizes the weighted sum that includes $r_c \cdot C_{f,u,c}$, so the optimizer pushes $C_{f,u,c}$ down to the largest binding constraint, i.e., the maximum of the two right-hand sides. Adds one continuous variable and two inequality constraints per (contract, flight, ULD type) triple.

9.2 Bilinear Rehandling Cost Linearization

The rehandling cost term contains $y^{c}_{f,u,k} \cdot y^{c'}_{g,u,k}$ (product of two binaries). Standard McCormick linearization: introduce auxiliary binary $\zeta_{f,g,u,k} \in \{0,1\}$ representing the product, with three constraints:

$$\zeta_{f,g,u,k} \;\leq\; y^{c}_{f,u,k} \tag{23}$$

$$\zeta_{f,g,u,k} \;\leq\; y^{c'}_{g,u,k} \tag{24}$$

$$\zeta_{f,g,u,k} \;\geq\; y^{c}_{f,u,k} + y^{c'}_{g,u,k} - 1 \tag{25}$$

Since ζ appears with positive coefficient in the objective (rehandling cost), the optimizer drives ζ down to 0 unless both y values force it to 1 — exactly the desired AND semantics. Adds one binary variable and three inequalities per consecutive arc pair per ULD type per shipment.

9.3 Big-M Tightening

The big-M constants in P.11, P.12, P.13, P.14 can be replaced with the tightest valid bound: e.g., the maximum possible time span across the planning horizon. Loose big-M values weaken the LP relaxation and slow solving. Practical tightening: use $M = T^{\text{dead}}_{k} - t^{\text{rdy}}_{k} + \epsilon$ on time-related constraints.

9.4 Spot Rate Per-Shipment Pre-Computation

The IATA weight-break rule (§4.6) is a min over tiers of piecewise linear cost. Since shipment chargeable weight cw_k is a parameter, $\text{cost}^{\text{spot}}(k, f)$ is fully pre-computed before the MILP is built. No linearization needed at solve time.

10 Re-ULDing — Operational Mechanics

This section motivates and defines re-ULDing in operational detail, which is the basis for the through-ULD compatibility parameter ψ and the rehandling cost ρ^{reULD}.

What re-ULDing is. At an intermediate (hub) airport on a multi-hop route, a built ULD arriving on flight f may need to be *broken down* — i.e., physically dismantled, with cargo removed piece-by-piece — and *rebuilt* into a different ULD for the onward flight g. The specific sequence:

1. ULD arrives on inbound flight f at the hub.
2. Ground handler unloads ULD from aircraft into the hub cargo terminal.
3. ULD is moved to the breakdown area.
4. Cargo is removed from the ULD; each AWB's pieces are extracted and tagged.
5. Cargo is held in the hub warehouse pending the onward flight g.
6. For onward flight g: an empty ULD from carrier(g)'s pool is delivered to build-up.
7. Cargo is restuffed into the new ULD, secured with netting, weighed.
8. Built ULD is moved to the aircraft loading area and loaded onto g.

When re-ULDing is required ($\psi = 0$).

- **Interline (different carriers):** carrier(f) $\neq$ carrier(g). Each airline maintains its own ULD pool; ULDs cannot transfer between pools. Cargo must leave airline f's ULD and enter airline g's ULD.
- **ULD type change:** if the inbound aircraft uses LD3s (wide-body belly) and the outbound aircraft uses PMC main-deck pallets (freighter), the ULD type itself must change. Cargo is re-stuffed into the new ULD type.
- **Hub lacks through-cargo facility for type u:** rare at major hubs (HKG, FRA, NRT all have through-cargo handling) but possible at smaller airports.
- **Forwarder elects re-ULD:** e.g., to re-sort cargo destined to different onward flights into different ULDs. Not modeled in MVP (deferred).

When through-ULD applies ($\psi = 1$).

- **Through-flight technical stop:** same flight number, same aircraft. The ULD stays in its aircraft position throughout the intermediate stop. The hub processing is limited to refueling and crew change. Example: 5Y9123 TPE→ANC→JFK, ULD loaded at TPE stays on the aircraft through ANC and continues to JFK.
- **Same-airline through-cargo at hub:** carrier(f) = carrier(g), the airline has a through-cargo agreement at the hub, and the ULD type is compatible with both aircraft. ULD is unloaded from f, moved as a complete unit through the hub through-cargo facility, and loaded onto g. Common at major hubs for same-airline connections (e.g., Cathay TPE→HKG→JFK on CX564 + CX880).

Cost and time impact.

- Through-ULD MCT: MCT^{thru} typically 1.5–4 hours depending on hub and aircraft turnaround.
- Re-ULD MCT: $\text{MCT}^{\text{reULD}}$ typically 6–12 hours including breakdown, hub storage, and rebuild for onward flight.
- Rehandling cost: ρ^{reULD} typically \$150–\$500 per ULD, depending on cargo characteristics and hub labor rates.
- Cargo risk: re-ULDing introduces handling-induced damage and piece-loss risk; valuable / fragile cargo prefers through-ULD where possible (modeled implicitly via the cost differential).

11 Open Items and Future Extensions

1. **Period-level commitment optimization (P.7).** MVP takes $\underline{M}_{c,u,t}$ from a separate upstream model. Future: joint optimization of period commitment allocation across the rolling horizon.

2. **Through-ULD as decision variable.** MVP makes ψ a pre-computed parameter. P1: promote to decision variable, allowing forwarder-elected re-ULDing at compatible hubs for cargo re-sorting.

3. **Probabilistic transit time objective.** MVP uses deterministic mean transit and a hard deadline. P1: extend to $\Pr(\text{arrival} \leq T_k^{\text{dead}}) \geq p$.

4. **Full 3D bin-packing within ULD.** MVP uses aggregate volume constraint (P.2) with $\eta = 0.97$ as a fill-rate proxy. P1: 3D bin-packing model for shipments with awkward piece dimensions.

5. **ULD availability at CFS.** MVP assumes empty ULDs are always delivered to CFS before cargo cutoff. P1: model ULD pool availability as a constraint.
6. **CFS dock door capacity.** MVP assumes infinite CFS throughput. P1: model dock door and labor capacity as a CFS-level constraint.
7. **Cargo type segregation within ULD.** MVP allows any compatible commodities to share a ULD. P1: hazmat compatibility matrix, perishable temperature segregation.
8. **Multiple ULD types per shipment.** MVP forces a shipment to one ULD type per flight. P1: allow splitting across LD3 + LD7 if economically preferred.
9. **Carrier alliance slot sharing.** MVP treats each contract independently. P1: model SkyTeam / Star Alliance / oneworld slot exchanges where ULD positions on partner carriers can substitute.
10. **Spot capacity availability uncertainty.** MVP treats spot rate as always available. Real spot capacity has day-of-departure availability risk. P1: model spot capacity as a stochastic constraint or with explicit booking rejection events.